

1. According to the [AirVisual mobile application](#) as show in the figure 1, you must design the diagram 1.1-1.2 ([low level design for programmers](#)). The color code of AQI level will be updated according to AQI ranks in figure 2. The application will send the current location as city name to retrieve the latest information including AQI, temperature, wind speed, and humidity from the [AirVisual webservice](#) as a backend API and this webservice queries the latest data from [AirVisual database server](#). Note. The webservice API and database servers are located in different servers.

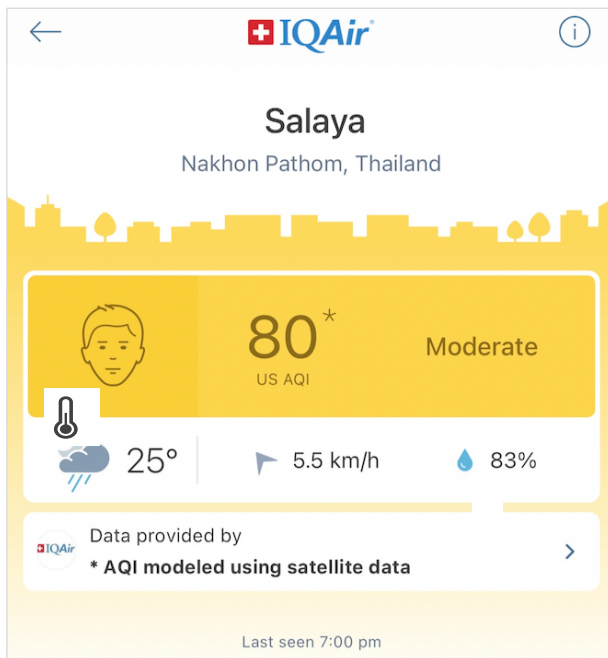


Figure 1

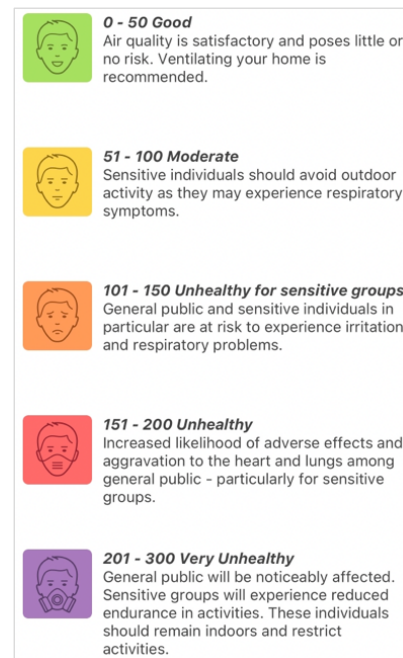


Figure 2

- 1.1. Draw class diagram for AirVisual mobile application, their web services and related relationship (2 marks)
 - You must consider functional dependencies for good software design
- 1.2. Draw sequence diagram (3 marks)
 - You must consider functional dependencies for good software design
- 1.3. What is the type of client-server architectures for this application? Draw software architecture diagram (2 marks)
- 1.4. Draw the C4 Model including the system context diagram, container diagram, and component diagram for each container. (3 marks)